Pre-course setup guide

Introduction to AI-Assisted Research Methods and Data Analysis

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Before you start

Please complete this setup **before the online setup clinic**, about one week before the course. It takes 45 to 60 minutes. If you get stuck, stop and bring your laptop to the clinic: we will fix it together.

You need a laptop on which you are allowed to install software. If you cannot install anything (for example a locked-down work laptop), skip to *No-install route* at the end.

Choose one language. The course runs in Python or R; both give identical results. If you have no preference, choose Python.

Step 1: Install the core tools

Tool	Where	Notes
VS Code	code.visualstudio.com	Accept the defaults
Quarto	quarto.org (Get Started)	Used on Day 4 to render the report
Zotero desktop	zotero.org/download	Install the Zotero Connector for your browser too
Git (optional)	git-scm.com	Not required for the course

Step 2: Install your language

Python route

1. Install Python 3.11 or newer from python.org (Windows: tick **Add python.exe to PATH**).
2. Open the course folder in VS Code (**File > Open Folder**).
3. Open the terminal (**Terminal > New Terminal**) and run:

```
python -m venv .venv
```

then activate it:

```
Windows:       .venv\Scripts\activate
macOS / Linux: source .venv/bin/activate
```

and install the packages:

```
pip install -r requirements.txt
```

R route

1. Install R 4.3 or newer from cran.r-project.org.
2. Open R (or RStudio / Positron) and run, from inside the course folder:

```
source("install_r_packages.R")
```

This installs the tidyverse, broom, knitr and IRkernel, and registers R so VS Code can find it.

Step 3: VS Code extensions

In VS Code, open **Extensions** (the four-squares icon) and install:

- **Python** and **Jupyter** (Microsoft)
- **GitHub Copilot** (GitHub)
- **Quarto** (Quarto)
- For R users: **R** (REditorSupport) is helpful but optional

Step 4: GitHub Copilot

1. Create a free account at github.com if you do not have one.
2. In VS Code, click the accounts icon (bottom left) and **sign in with GitHub**.
3. Copilot's free plan is enough for the course. Students and teachers can apply for more through GitHub Education.
4. Check your institution's policy on AI coding assistants. If you already pay for another assistant (Gemini Code Assist, Claude, ChatGPT, Cursor) you may use it instead.

Step 5: Run the setup check

1. In VS Code, **File > Open Folder** and choose the whole course folder.
2. Open `notebooks/python/00_setup_and_orientation.ipynb` (or `notebooks/r/...` for R).
3. Click **Select Kernel** (top right). Python: choose the `.venv` environment. R: choose **Jupyter Kernel > R**.
4. Click **Run All**.
5. You are ready when the last cell prints **Setup check passed**.

Step 6: Accounts for Day 1

- **Zotero:** create a free account at zotero.org (optional, for syncing).
- **NotebookLM:** sign in at notebooklm.google with a Google account, **only if your institution allows it**. A printed alternative is provided.

No-install route: Google Colab

1. Upload the whole course folder to your Google Drive.
2. Open `notebooks/python/00_setup_and_orientation.ipynb` from Drive with **Google Colaboratory**.

3. Add and run this cell at the top, changing the path to where you put the folder:

```
from google.colab import drive

drive.mount("/content/drive")

%cd "/content/drive/MyDrive/AI-Assisted Research Methods/notebooks/python"

!pip install -q pyreadstat
```

4. Then **Runtime > Run all**. Colab has Gemini built in as its AI assistant. Colab also offers an R runtime (**Runtime > Change runtime type**).

What to bring

- Your laptop and charger
- Your GitHub and Google sign-in details
- One clinical question you have wondered about